

THE APÉRY GCD: A DENSITY-ONE THEOREM AND THE POINTWISE ENDPOINT

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ABSTRACT. Let a_n and b_n be the two standard solutions of the Apéry recurrence for $\zeta(3)$. Put

$$d_n = \text{lcm}(1, \dots, n)^3, \quad G_n = \text{gcd}(d_n a_n, d_n b_n).$$

We prove that for every $\varepsilon > 0$ the set of $n \leq X$ for which $\log G_n > \varepsilon n$ has cardinality $o(X)$. More precisely, on every dyadic shell $(N, 2N]$ the average of $\log G_n$ is $O(N^{2/3} \log N)$. For the pointwise problem we identify the exact radical middle prime channel, prove a reflection functional equation, and construct fixed gap-eliminant polynomials U_g . In the only delicate range we lower the pointwise problem to a depth-2 pair gate and then to a single-arc exact-collision conjecture. We prove complementary cross-arc and structured-resonance results, give two algebraic descriptions of the zero sets, and record barrier certificates showing why horizontal equidistribution does not resolve the remaining scale-zero collision problem.

1. STATEMENT

Apéry's construction [3] gives two $\zeta(3)$ solutions b_n and a_n of

$$(n+1)^3 u_{n+1} = (2n+1)(17n^2 + 17n + 5)u_n - n^3 u_{n-1},$$

with $(b_0, b_1) = (1, 5)$, where the b_n are integers, and $(a_0, a_1) = (0, 6)$, where the a_n are rational and

$$v_p(a_n) \geq -3 \lfloor \log_p n \rfloor.$$

Set

$$d_n = \text{lcm}(1, \dots, n)^3, \quad G_n = \text{gcd}(d_n a_n, d_n b_n).$$

Theorem A. *For every $\varepsilon > 0$,*

$$\#\{n \leq X : \log G_n > \varepsilon n\} = o(X).$$

Quantitatively, on every dyadic shell,

$$\frac{1}{N} \sum_{N < n \leq 2N} \log G_n = O(N^{2/3} \log N).$$

Write

$$P(t) = (2t+1)(17t^2 + 17t + 5)$$

throughout. Unless stated otherwise, p always denotes a prime at least 7. The primes 2, 3, 5 contribute $O(\log^2 n)$ to $\log G_n$, since

$$v_p(G_n) \leq v_p(d_n) + O(\log_p n) = O(\log n),$$

and they are absorbed into the small-prime estimate.

Remark 1 (Provenance and status). This proof chain was assembled on July 19, 2026, by extracting and streamlining `Q3.2_paper_draft.md`. Every lemma used below is included, and a summary of the accompanying exact verification appears in the appendix.

2. PRELIMINARIES

2.1. Wronskian. Define

$$W_n := a_{n+1}b_n - a_nb_{n+1}.$$

Then

$$W_n = \frac{6}{(n+1)^3}.$$

Indeed, $W_0 = 6$, and the recurrence gives $W_n = \frac{n^3}{(n+1)^3}W_{n-1}$. In particular, $W_n \neq 0$, and for $\sqrt{n} < p \leq n$, with $p \geq 7$,

$$v_p(G_n) \leq 6.$$

Proof. The integer G_n divides both d_na_n and d_nb_n , hence it divides the integer combination

$$\begin{aligned} & d_{n+1}b_{n+1}(d_na_n) - d_{n+1}a_{n+1}(d_nb_n) \\ &= -d_nd_{n+1}W_n \\ &= -\frac{6d_nd_{n+1}}{(n+1)^3}. \end{aligned}$$

For $\sqrt{n} < p \leq n$, one has $v_p(d_n), v_p(d_{n+1}) \leq 3$ and $p \nmid 6$. Therefore

$$v_p(G_n) \leq 6 + v_p(6) - 3v_p(n+1) \leq 6.$$

□

Remark 2 (Audit correction). Any absolute constant depth suffices for the averaging argument. The bound 6 above replaces an earlier unproved bound 4 identified during the hostile audit.

2.2. Cleared continuants. Let

$$N_0 = 0, \quad N_1 = 1, \quad N_{r+1}(x) = P(x+r)N_r(x) - (x+r)^6N_{r-1}(x) \in \mathbb{Z}[x].$$

Then $\deg N_r = 3(r-1)$ for $r \geq 1$. Every solution of the local cleared recurrence at base x satisfies

$$(1.2) \quad \begin{aligned} Q_h(x)y_h &= N_h(x)y_1 - (x+1)^3N_{h-1}(x+1)y_0, \\ Q_h(x) &= \prod_{i=2}^h (x+i)^3. \end{aligned}$$

2.3. Lucas congruence. If $n = \sum_i d_i p^i$ is the base- p expansion of n , then

$$b_n \equiv \prod_i b_{d_i} \pmod{p}.$$

This is Gessel's Lucas congruence [24]; related divisibility and Lucas phenomena for Apéry-like sequences are developed in [34, 33].

3. FIVE CONTINUANT LEMMAS

Lemma 1 (No consecutive zeros; both solutions). *For $p \geq 7$ and $0 \leq r \leq p-2$, it is not the case that both $b_r \equiv 0 \pmod{p}$ and $b_{r+1} \equiv 0 \pmod{p}$. For $1 \leq q \leq p-2$, it is not the case that both $a_q \equiv 0 \pmod{p}$ and $a_{q+1} \equiv 0 \pmod{p}$; here a_q is p -integral because $q < p$.*

Proof. Two consecutive zeros back-propagate through the recurrence, because all coefficients m^3 with $m < p$ are invertible. For b this reaches $b_0 = 1$, a contradiction. For a it reaches $a_1 = 6$, and $p \nmid 6$. $\square$

Lemma 2 (Evaluation identities). *For every $r \geq 1$,*

- (i) $N_r(-1) = ((r-1)!)^3 b_{r-1}$;
- (ii) $N_r(-x-r) = (-1)^{r-1} N_r(x-1)$, and consequently

$$N_r(-r) = (-1)^{r-1} ((r-1)!)^3 b_{r-1}.$$

Proof. For the first identity, $v_r := N_r(-1)$ satisfies $v_1 = 1$, $v_2 = P(0) = 5$, and

$$v_{r+1} = P(r-1)v_r - (r-1)^6 v_{r-1}.$$

Substituting $v_r = ((r-1)!)^3 b_{r-1}$ reduces this to the Apéry recurrence at $n = r-1$.

For the second identity, under $x \mapsto -x-r$ each diagonal entry of the defining tridiagonal matrix transforms by $P(-1-t) = -P(t)$, and each off-diagonal cube is negated, in reversed order. Simultaneous row-and-column reversal preserves the determinant, while the negation contributes $(-1)^{r-1}$ because the matrix has size $r-1$. $\square$

Lemma 3 (Continuant lemma). *For $p \geq 7$ and $1 \leq h < p$, one has $N_h \not\equiv 0 \pmod{p}$. For $p = 5$ the same assertion follows by a direct check for $h < 5$.*

Proof. Suppose that $N_h \equiv 0 \pmod{p}$. Lemma 2(i) gives $b_{h-1} \equiv 0 \pmod{p}$. The defining recurrence, evaluated at $x = 1-h$, gives

$$N_h(1-h) = P(0)N_{h-1}(1-h) = 5N_{h-1}(1-h).$$

Since $p \neq 5$, it follows that $N_{h-1}(1-h) = N_{h-1}(-(h-1)) \equiv 0 \pmod{p}$. By Lemma 2(ii), $b_{h-2} \equiv 0 \pmod{p}$, contrary to Lemma 1. $\square$

Lemma 4 (Pair certificate). *Let y be any solution of the Apéry recurrence modulo p on a window $[m, m+h]$, where $m+h < p$. If*

$$y_m \equiv y_{m+h} \equiv 0 \pmod{p} \quad \text{and} \quad y_{m+1} \not\equiv 0 \pmod{p},$$

then $N_h(m) \equiv 0 \pmod{p}$.

Proof. Formula (1.2) at base $x = m$, with $y_0 = 0$, gives

$$Q_h(m)y_{m+h} = N_h(m)y_{m+1}.$$

All factors $m+i$ for $2 \leq i \leq h$ lie in $(m, m+h] \subseteq (0, p)$, so $Q_h(m)$ is a unit. The two hypotheses on y_{m+h} and y_{m+1} force $N_h(m) \equiv 0 \pmod{p}$. $\square$

Lemma 5 (Zero-fiber bounds). *For $p \geq 7$,*

$$\mathcal{Z}_b(p) := \{0 \leq r < p : b_r \equiv 0 \pmod{p}\},$$

$$Z_b(p) := |\mathcal{Z}_b(p)| \leq 3p^{2/3} + O(1),$$

and

$$\mathcal{Z}_a(p) := \{1 \leq q < p : a_q \equiv 0 \pmod{p}\},$$

$$Z_a(p) := |\mathcal{Z}_a(p)| \leq 3p^{2/3} + O(1).$$

Proof. Let $r_1 < \dots < r_z$ be the zeros and set $h_i = r_{i+1} - r_i$, so that $\sum_i h_i < p$. By Lemmas 1 and 4, every consecutive pair with $h_i \leq K$ makes r_i a root of $\bar{N}_{h_i} \neq 0$; the nonvanishing follows from Lemma 3. For a fixed h the bases are distinct. Since $\deg N_h = 3(h-1)$, at most

$$\sum_{h \leq K} 3(h-1) < \frac{3}{2}K^2$$

pairs have gap at most K , while fewer than p/K have gap greater than K . Taking $K = p^{1/3}$ proves the bound for $Z_b(p)$. The a -case is identical; the zero $a_0 = 0$ is excluded from the count, and its windows start at $q \geq 1$. $\square$

4. THE SUPPORT LAW

Write $\tilde{a}_n = p^3 a_n$ where relevant; this quantity is p -integral for $n < p^2$. To make the binomial notation used below explicit, set

$$H_N^{(3)} = \sum_{j=1}^N \frac{1}{j^3}, \quad T(N, K) = \binom{N}{K}^2 \binom{N+K}{K}^2,$$

and

$$E(N, K) = H_N^{(3)} + \sum_{j=1}^K \frac{(-1)^{j-1}}{2j^3 \binom{N}{j} \binom{N+j}{j}}.$$

The second Apéry solution has the standard representation

$$a_N = \sum_{K=0}^N T(N, K) E(N, K).$$

For background on congruences for these and closely related Apéry sums, see [36, 49, 50].

Lemma 6 (The congruence $\alpha_p \equiv 6$). *For $p \geq 7$,*

$$p^3 a_p \equiv 6 \pmod{p}.$$

Proof. In the binomial sum $a_p = \sum_{k=0}^p T(p, k)E(p, k)$, the p^3 -pole survives only at $k = 0$ and $k = p$. For $1 \leq k \leq p-1$, one has $v_p\left(\binom{p}{k}^2\right) \geq 2$. After multiplication by p^3 , the $H_p^{(3)}$ part therefore has valuation at least 2, while the inner sum, whose p -adic valuation is at least -1 , has valuation at least 4. Thus all these terms vanish modulo p . The $k = 0$ term contributes

$$p^3 H_p^{(3)} = p^3 H_{p-1}^{(3)} + 1 \equiv 1 \pmod{p}.$$

The $k = p$ term contributes

$$\binom{2p}{p}^2 \left(1 + \frac{1}{4}\right) \equiv 4 \cdot \frac{5}{4} = 5 \pmod{p}.$$

Here Lucas gives $\binom{2p}{p} \equiv 2 \pmod{p}$; the $j = p$ inner term gives

$$\frac{p^3}{2p^3 \binom{2p}{p}} \equiv \frac{1}{4} \pmod{p},$$

and the inner terms with $j < p$ have pole order at most 1. The total is $1 + 5 = 6$. $\square$

Lemma 7 (Boundary lemma). *For $1 \leq q < p$ and $p \geq 7$,*

$$p^3 a_{qp} \equiv a_q \pmod{p}.$$

Kummer bookkeeping. In the sum for a_{qp} , $T(qp, K)$ is a p -unit if and only if $K = \ell p$ with $q + \ell < p$. Indeed, by Kummer, any $s > 0$ in $K = \ell p + s$ forces borrows. Together with the carry indicator of $\binom{qp+K}{K}$, this gives total valuation at least 2. The bound $v_p(E(qp, K)) \geq -3$ shows that non-unit terms die after multiplication by p^3 ; termwise, the inner terms with $p \nmid j$ have total valuation at least 3, and the terms $j = tp$ have total valuation at least 1.

On the surviving terms, Lucas gives $T(qp, \ell p) \equiv T(q, \ell) \pmod{p}$; moreover,

$$p^3 H_{qp}^{(3)} \equiv H_q^{(3)} \pmod{p},$$

because only the multiples of p survive, and the inner sum reduces index by index with $(-1)^{tp-1} = (-1)^{t-1}$. This yields

$$\sum_{\ell: q+\ell < p} T(q, \ell) E(q, \ell).$$

Restoring the terms with $q + \ell \geq p$ costs nothing: there $v_p(T(q, \ell)) = 2$ from one carry, squared, while $v_p(E(q, \ell)) \geq -1$. $\square$

Lemma 8 (Block law in the r -direction). *For $0 \leq r < p$ and $1 \leq q < p$,*

$$\tilde{a}_{qp+r} \equiv \tilde{a}_{qp} b_r \pmod{p}.$$

Proof. Multiply the recurrence at $n = qp + r$ by p^3 . For $1 \leq r \leq p-2$, the coefficients reduce to those in the b -recurrence. At the entry $r = 0$, the

backward term carries $(qp)^3 = q^3 p^3$ against the p -integral number $\tilde{a}_{qp-1}$, and hence vanishes modulo p ; also $(qp+1)^3 \equiv 1 \pmod{p}$, so

$$\tilde{a}_{qp+1} \equiv P(0)\tilde{a}_{qp} = 5\tilde{a}_{qp} \pmod{p}.$$

Thus both sides satisfy the same nondegenerate recurrence with initial data

$$\tilde{a}_{qp}(1, 5) = \tilde{a}_{qp}(b_0, b_1).$$

□

Theorem B (Support law). *Let $p \geq 7$ and $\sqrt{n} < p \leq n$. Write $n = qp + r$, where $1 \leq q < p$ and $0 \leq r < p$. Then*

$$p \mid G_n \iff a_q b_r \equiv 0 \pmod{p}.$$

Proof. In this range $v_p(d_n) = 3$, and $v_p(b_n) \geq 0$. Hence

$$p \mid G_n \iff v_p(a_n) \geq -2 \iff \tilde{a}_n = p^3 a_n \equiv 0 \pmod{p}.$$

By Lemmas 7 and 8,

$$\tilde{a}_n \equiv \tilde{a}_{qp} b_r \equiv a_q b_r \pmod{p}.$$

□

Remark 3 (Historical correction). The earlier one-channel claim “ $p \mid G_n \Rightarrow b_r \equiv 0 \pmod{p}$ ” is false outside $n/2 < p < n$: the hostile audit found the counterexamples $13 \mid G_{26}$ and $13 \mid G_{27}$. In the top range $q = 1$, however, $a_1 = 6$ is a unit, and the old claim is recovered.

5. PROOF OF THEOREM A

Proof of Theorem A. Fix the shell $(N, 2N]$. For n in this shell, abbreviate $v_p = v_p(G_n)$ and write

$$\log G_n = \underbrace{\sum_{p \leq \sqrt{n}} v_p \log p}_{S_1(n)} + \underbrace{\sum_{\sqrt{n} < p \leq n} v_p \log p}_{S_2(n)}.$$

Small primes. For every $p \leq \sqrt{n}$,

$$v_p(G_n) \leq v_p(d_n) + \max(0, v_p(a_n)) = O(\log_p n).$$

Consequently,

$$S_1(n) = O(\pi(\sqrt{n}) \log n) = O(\sqrt{n}).$$

Large primes. By Theorem B, a prime p is bad if and only if $b_r \equiv 0 \pmod{p}$ in the lower channel or $a_q \equiv 0 \pmod{p}$ in the companion channel. By the Wronskian depth estimate, each bad prime costs at most $6 \log(2N)$.

For the lower channel, fix p . The shell integers with $n \bmod p$ in the zero fiber counted by $Z_b(p)$ number at most

$$Z_b(p) \left(\frac{N}{p} + O(1) \right).$$

The endpoint term is $O(Z_b(p))$ for each prime. Lemma 5 therefore gives

$$\begin{aligned} \sum_{p \leq 2N} Z_b(p) \left(\frac{N}{p} + O(1) \right) &\ll N \sum_{p \leq 2N} p^{-1/3} + \sum_{p \leq 2N} p^{2/3} \\ &\ll N^{5/3}. \end{aligned}$$

For the companion channel, fix $q \geq 1$. The congruence $a_q \equiv 0 \pmod{p}$ means that p divides the fixed nonzero integer $\text{num}(a_q)$. This integer is nonzero because $a_q > 0$ for $q \geq 1$, and

$$\log |\text{num}(a_q)| = O(q).$$

It therefore has at most $O(q/\log N)$ prime divisors exceeding $\sqrt{N}$. Each such pair (q, p) poisons at most $p \leq 2N/q$ shell integers, namely those for which $\lfloor n/p \rfloor = q$. The total number of incidences is

$$\sum_{q \leq 2\sqrt{2N}} \frac{2N}{q} O\left(\frac{q}{\log N}\right) = O\left(\frac{N^{3/2}}{\log N}\right).$$

Assembly. Summing both channels over the shell and multiplying by the depth cost gives

$$\begin{aligned} \sum_{N < n \leq 2N} S_2(n) &\ll \left(N^{5/3} + \frac{N^{3/2}}{\log N} \right) \log N \\ &\ll N^{5/3} \log N. \end{aligned}$$

Together with the small-prime estimate, this shows that the shell average of $\log G_n$ is $O(N^{2/3} \log N)$. Markov's inequality on the shell now yields

$$\begin{aligned} \#\{n \in (N, 2N] : \log G_n > \varepsilon n\} &\leq \#\{n \in (N, 2N] : \log G_n > \varepsilon N\} \\ &\ll_{\varepsilon} N^{2/3} \log N. \end{aligned}$$

Summing over dyadic shells up to X gives $o(X)$. □

6. THE POINTWISE REDUCTION

Put

$$b_m = \sum_{k=0}^m \binom{m}{k}^2 \binom{m+k}{k}^2, \quad Z_p = \{0 \leq r < p : b_r \equiv 0 \pmod{p}\},$$

and write $e(t) = \exp(2\pi i t)$ and $\|t\| = \text{dist}(t, \mathbb{Z})$.

6.1. Reflection.

Lemma 9 (Reflection functional equation). *For every prime p and every $0 \leq n \leq p-1$,*

$$b_{p-1-n} \equiv b_n \pmod{p}.$$

Proof. For $0 \leq k < p$, the product formula for binomial coefficients gives

$$\binom{p-1-m}{k} \equiv (-1)^k \binom{m+k}{k} \pmod{p}.$$

Apply this first with $m = n$ and then with $m = n - k$. Thus

$$\begin{aligned} \binom{p-1-n}{k} &\equiv (-1)^k \binom{n+k}{k}, \\ \binom{p-1-n+k}{k} &\equiv (-1)^k \binom{n}{k} \pmod{p}. \end{aligned}$$

Both Apéry sums may be extended to $0 \leq k < p$: outside the original summation range the corresponding binomial factor is zero modulo p . The two displayed congruences therefore identify their summands term by term, after squaring, and summation proves the result. $\square$

This direct proof is Lemma 6.2 of Malik–Straub [34]; see also the broader congruence work of Beukers and Coster [4, 14]. The identity was also checked with no failure for all primes $p \leq 12809$; the termwise congruence was checked for $p \leq 809$.

6.2. The common lower channel. Let $\sqrt{n} < p \leq n$ and write $n = qp + r$, with $q = \lfloor n/p \rfloor$ and $0 \leq r < p$. The support law of Theorem B and the Gessel–Coster Lucas congruence give two different two-channel decompositions:

$$\begin{aligned} (1) \quad p \mid G_n &\iff a_q b_r \equiv 0 \pmod{p}, \\ (2) \quad p \mid b_n &\iff b_q b_r \equiv 0 \pmod{p}. \end{aligned}$$

The second equivalence follows from $b_{qp+r} \equiv b_q b_r \pmod{p}$ [24, 14]. The two decompositions share exactly the condition $r \in Z_p$.

Lemma 10 (Both digit channels are pointwise-free). *For every n ,*

$$\begin{aligned} \#\{p \in (\sqrt{n}, n] : a_{\lfloor n/p \rfloor} \equiv 0 \pmod{p}\} &\ll n^{2/3}, \\ \#\{p \in (\sqrt{n}, n] : \lfloor n/p \rfloor \in Z_p\} &\ll n^{2/3}. \end{aligned}$$

Proof. For fixed q , every prime under consideration lies in

$$I_q(n) = \left(\frac{n}{q+1}, \frac{n}{q} \right], \quad |I_q(n)| < \frac{n}{q^2}.$$

In the first line it divides the fixed nonzero integer $\text{num}(a_q)$; in the second it divides the fixed positive integer b_q . The standard growth estimates

$\log |\text{num}(a_q)|, \log b_q = O(q)$ show that either integer has $O(q/\log n)$ prime factors exceeding $\sqrt{n}$. Hence each $q \leq \sqrt{n}$ contributes at most

$$\min \left(\frac{n}{q^2} + 1, \frac{Cq}{\log n} \right).$$

Splitting the sum at $q = n^{1/3}$ proves both assertions. $\square$

Define the radical lower-channel mass

$$R_b(n) = \sum_{\substack{\sqrt{n} < p \leq n \\ n \bmod p \in Z_p}} \log p$$

and the radical middle part of b_n by

$$M_{\text{rad}}(n) = \sum_{\substack{\sqrt{n} < p \leq n \\ p | b_n}} \log p.$$

Theorem C (Exact pointwise endpoint). *As $n \rightarrow \infty$,*

$$(3) \quad R_b(n) \leq \log G_n,$$

$$(4) \quad \log G_n \leq 6R_b(n) + O(n^{2/3} \log n + \sqrt{n}),$$

$$(5) \quad M_{\text{rad}}(n) = R_b(n) + O(n^{2/3} \log n).$$

Consequently,

$$\log G_n = o(n) \quad \text{for every } n \iff M_{\text{rad}}(n) = o(n) \quad \text{for every } n.$$

Proof. If $n \bmod p \in Z_p$, then (1) gives $p \mid G_n$, which proves (3). Conversely, every prime in the large range dividing G_n belongs either to the common lower channel or to the first digit channel in Lemma 10; each has depth at most 6. The small-prime estimate in Section 5 is $O(\sqrt{n})$, proving (4). Finally, (2) and the second line of Lemma 10 give (5). $\square$

Conjecture 1 (Full-multiplicity middle part). *The stronger full-multiplicity statement requested in the original program is*

$$(T) \quad \log \prod_{\substack{\sqrt{n} < p \leq n \\ p | b_n}} p^{v_p(b_n)} = o(n).$$

It implies the pointwise theorem, but the converse is not proved: the gcd sees each middle prime only to depth at most 6. Thus Theorem C, not (T) with full multiplicity, is the unconditional equivalence.

Remark 4 (Rhin–Viola correction). The Rhin–Viola irrationality measure [40] gives

$$\limsup_{n \rightarrow \infty} \frac{\log G_n}{n} \leq 3 + \log(17 + 12\sqrt{2}) - \frac{2 \log(17 + 12\sqrt{2})}{5.51389 \dots} = 5.24673 \dots$$

The earlier value, $2.24672 \dots$, omitted the contribution $\log d_n/n \rightarrow 3$ and is retracted.

6.3. The in-range window. For a dyadic parameter X , put

$$K_X(y) = \#\{p \in (X, 2X] : y \bmod p \in Z_p\}.$$

Remark 5 (The required range). Fix $A > 0$. A full block $(X, 2X]$ is delicate only when

$$(X, 2X] \subset \left(\frac{n}{(\log n)^A}, n \right].$$

After increasing A harmlessly, all such blocks are covered by the working window

$$(6.1) \quad \mathcal{J}_X = \{y \in \mathbb{Z} : 2X \leq y \leq XL\}, \quad L = (\log X)^A.$$

If $X \leq n/(\log n)^A$, the trivial estimate $K_X(n) \leq \pi(2X)$ closes the block after multiplication by its logarithmic weight. Thus no estimate at $n = O(1)$ is used.

Lemma 11 (Intercept vacuity). *Suppose $p, p' \in (X, 2X]$ are distinct and two incidences lie on the affine line*

$$r = up + v, \quad r' = up' + v.$$

If their unwrapped phases are X^{-1} -close at n , then

$$(6.2) \quad |p - p'| \leq \frac{pp'}{X|n - v|} \leq \frac{4X}{|n - v|}.$$

Consequently, for fixed B , if $|v| \ll (\log X)^B$ and $n \geq 2X$, then $|p - p'| \leq 2$ for all sufficiently large X . A line carrying T incidences contributes only $O(T)$ close pairs, independently of any ordinarity assertion.

Proof. The pair intercept is

$$\frac{rp' - r'p}{p' - p} = v,$$

and on the lift determined by the line,

$$\left| \frac{r - n}{p} - \frac{r' - n}{p'} \right| = \frac{|n - v| |p - p'|}{pp'}.$$

The close-pair condition and $pp' \leq 4X^2$ give (6.2). Its right-hand side is $2 + o(1)$ in the stated range; integrality gives the assertion. Each prime has only $O(1)$ possible partners at those gaps. $\square$

The central Apéry zero, when it occurs, has intercept $-1/2$. Lemma 11 therefore shows that density-one ordinarity for the non-CM form 8.4.a.a is irrelevant to the in-range problem. The analogous forced central line for the $\zeta(2)$ Apéry sequence is likewise an out-of-range obstruction only.

Conjecture 2 (The first shifted-prime track). *For every fixed $A > 0$, uniformly for $y \in \mathcal{J}_X$,*

$$\#\{p \in (X, 2X] : 0 \leq y - p < p, p \mid b_{y-p}\} = o\left(\frac{\pi(X)}{L}\right).$$

This is the slope-1 member of the terminal single-arc family below. It is a shifted-prime divisor problem for a fixed holonomic sequence, in the circle of pointwise questions represented by Hooley and by Murty–Rosen–Silverman [28, 37]. It is not used as a substitute for the all-slope collision conjecture.

Proposition 1 (Dyadic maximum criterion). *Suppose that, for some fixed $\delta > 0$ and every sufficiently large X ,*

$$\max_{m < X^2} K_X(m) \ll X^{1-\delta}.$$

Then $\log G_n = o(n)$ for every n . More generally, the same conclusion holds if $|E_X| \ll X^{1-\delta}$ and the displayed maximum bound is known for

$$K_X^G(m) = \#\{p \in (X, 2X] \setminus E_X : m \bmod p \in Z_p\}.$$

Proof. The blocks with $X < 2\sqrt{n}$ cost $O(\sqrt{n} \log n)$. On every remaining block, $n < X^2$, and the assumed estimate gives $O(X^{1-\delta} \log(2X))$. The dyadic sum is dominated by the top block and is $O(n^{1-\delta} \log n) = o(n)$. The same calculation bounds the discarded primes without imposing any condition on their residues. Theorem C completes the proof. $\square$

6.4. The depth-2 pair gate. Write

$$\mathcal{M}_X = \sum_{X < p \leq 2X} |Z_p|.$$

Conjecture 3 (Mass conjecture (C1)). *For every $\varepsilon > 0$,*

$$(C1) \quad \mathcal{M}_X \ll_{\varepsilon} X^{1+\varepsilon}.$$

The proved bound in Lemma 5 is $\mathcal{M}_X \ll X^{5/3+o(1)}$. Computations instead suggest about one root per prime on average: after grouping noncentral roots into reflection pairs, the pair count is close to a Poisson variable of mean $1/2$. Thus $\mathbb{E}|Z_p| \approx 1$, not $1/8$. The central fixed point prevents an unconditional parity assertion for $|Z_p|$.

For a set E_X of exceptional block primes define

$$(6.3) \quad P_{2,X}^G = \sum_{y \in \mathcal{J}_X} \binom{K_X^G(y)}{2}.$$

We write $P_{2,X}$ for the same quantity with $E_X = \emptyset$.

Proposition 2 (Pair rigidity and the depth-2 gate). *For fixed A and sufficiently large X , a pair of distinct block primes together with one zero class at each prime has at most one landing in $\mathcal{J}_X$. Moreover,*

$$(6.4) \quad \sup_{y \in \mathcal{J}_X} K_X^G(y) \leq 1 + \sqrt{2P_{2,X}^G}.$$

Thus, for any $\delta > 0$,

$$(6.5) \quad P_{2,X}^G \ll X^{2-2\delta} \implies \sup_{y \in \mathcal{J}_X} K_X^G(y) \ll X^{1-\delta}.$$

Together with $|E_X| \ll X^{1-\delta}$ at every dyadic scale, this implies the pointwise conclusion.

Proof. Two simultaneous zero-class congruences have period pq . Since $p, q > X$, one has $pq > X^2 > XL = \max \mathcal{J}_X$ for sufficiently large X , so two solutions cannot lie in the window. For each fixed y , $(K_X^G(y)) \leq P_{2,X}^G$, which gives (6.4). The last assertion is Proposition 1. $\square$

Conjecture 4 (Exceptional-tolerant pair gate $(P_2)^G$). *For every fixed $A > 0$ there are $\delta > 0$ and sets E_X , chosen independently of y , such that*

$$(P_2^G) \quad |E_X| \ll X^{1-\delta}, \quad P_{2,X}^G \ll X^{2-2\delta}.$$

Under (C1), Markov's inequality permits the canonical choice $E_X = \{p : |Z_p| > X^\varepsilon\}$, for which $|E_X| \leq X^{1-\varepsilon+o(1)}$. Also under (C1), the existence of some power saving in (P_2^G) is equivalent, after renaming the exponent, to a power-saving bound for $\sup K_X^G$. Indeed,

$$\sum_{y \in \mathcal{J}_X} K_X^G(y) \leq L\mathcal{M}_X, \quad P_{2,X}^G \leq \frac{1}{2}(\sup K_X^G) \sum_y K_X^G(y).$$

This is the exact depth-2 target. It requires neither a third moment nor an input of Elliott–Halberstam strength.

6.5. Single arcs and exact collisions. For $1 \leq a \leq L$ and $y \in \mathcal{J}_X$, define

$$S_{a,X}(y) = \{p \in (X, 2X] : 0 \leq y - ap < p, \ y - ap \in Z_p\},$$

$$R_{a,X}(y) = |S_{a,X}(y)|.$$

The digit restriction is part of the definition. It gives

$$(6.6) \quad K_X(y) = \sum_{1 \leq a \leq L} R_{a,X}(y), \quad \sum_{y \in \mathcal{J}_X} R_{a,X}(y) \leq \mathcal{M}_X.$$

Cauchy's inequality and (6.3) yield

$$(6.7) \quad P_{2,X} \leq \frac{1}{2} \sum_y K_X(y)^2 \leq \frac{L^2}{2} \max_{a \leq L} \sum_{y \in \mathcal{J}_X} R_{a,X}(y)^2.$$

The Gessel–Coster Lucas congruence gives, for $p \in S_{a,X}(y)$,

$$b_y = b_{ap+(y-ap)} \equiv b_a b_{y-ap} \equiv 0 \pmod{p}.$$

Consequently

$$(6.8) \quad R_{a,X}(y) \leq \omega_{(X, 2X]}(b_y),$$

where $\omega_I(m)$ counts distinct prime divisors of m in I .

Conjecture 5 (Single-arc cap (SA)). *For every fixed $A > 0$ there is $\eta > 0$ such that, uniformly for $a \leq L$ and $y \in \mathcal{J}_X$,*

$$(SA_\eta) \quad R_{a,X}(y) \ll_A X^{1-\eta}.$$

Conjecture 6 (Middle-prime cap (Ω)). *For every fixed $A > 0$ there is $\eta > 0$ such that, uniformly for $y \in \mathcal{J}_X$,*

$$(\Omega_\eta) \quad \omega_{(X, 2X]}(b_y) \ll_A X^{1-\eta}.$$

In the existential “some positive power” sense, Conjectures 5 and 6 are equivalent after renaming η . One direction is (6.8). Conversely, Lucas splits a prime counted by $\omega_{(X, 2X]}(b_y)$ between the lower digit and the condition $p \mid b_{\lfloor y/p \rfloor}$. The latter channel is $O(y^{2/3})$ by Lemma 10, while the lower channel is at most $L \max_a R_{a,X}(y)$.

Define the ordered exact-collision count

$$(6.9) \quad N_{\text{coll},a}(X) = \#\{(p, z; q, w) : p \neq q, p, q \in (X, 2X], z \in Z_p, w \in Z_q, ap + z = aq + w \in \mathcal{J}_X\}.$$

Opening the square gives the exact identity

$$(6.10) \quad \sum_{y \in \mathcal{J}_X} R_{a,X}(y)^2 = \sum_{y \in \mathcal{J}_X} R_{a,X}(y) + N_{\text{coll},a}(X).$$

Conjecture 7 (Terminal exact-collision conjecture). *For every fixed $A > 0$ there is $\eta > 0$ such that*

$$(\text{NC}_\eta) \quad \max_{1 \leq a \leq L} N_{\text{coll},a}(X) \ll_A X^{2-\eta}.$$

Theorem D (Exceptional-tolerant reduction to exact collisions). *Conjecture 7 implies the pointwise conclusion $\log G_n = o(n)$. Under Conjecture 3, the three properties (SA), (Ω) , and (NC) are equivalent in the sense that each asserts the existence of some fixed positive power saving; their exponents need not be the same. They are also equivalent, in this sense, to the power-saving target for $\sup K_X$ and to $(P_2)^G$.*

Proof. From (6.9), for each a, y ,

$$R_{a,X}(y)(R_{a,X}(y) - 1) \leq N_{\text{coll},a}(X).$$

Thus (NC_η) gives $R_{a,X}(y) \ll X^{1-\eta/2}$. Conversely, (SA_η) and (C1) give

$$N_{\text{coll},a}(X) \leq \left(\sup_y R_{a,X}(y)\right) \sum_y R_{a,X}(y) \ll X^{2-\eta+o(1)}.$$

This proves the asserted equivalence, together with the Lucas-channel comparison above. Finally, (6.7), (6.10), and the proved estimate $\mathcal{M}_X \ll X^{5/3+o(1)}$ give a positive power saving for $P_{2,X}$. Proposition 2 and then Proposition 1 prove the pointwise conclusion. $\square$

Remark 6 (No exceptional- y loophole). An “off a sparse set of y ” formulation is insufficient for a pointwise theorem unless the exceptional fibers receive their own collision bound: one exceptional y may carry a fiber of size $X^{1-o(1)}$. The uniform formulation (SA_η) , or equivalently the global moment formulation (NC_η) , is the precise terminal statement used here.

Remark 7 (Numerical and structural evidence for the terminal conjecture). Direct computation supports (NC_η) . With G_n computed exactly for all $n \leq 15000$, the extreme values obey the Poisson-independence law

$$\max_{n \leq N} \log G_n \approx 0.85 (\log N)^2,$$

a single constant across three orders of magnitude, so $\log G_n/n = O((\log n)^2/n)$ with no phase transition. The two channels of a hypothetical counterexample are both closed numerically. The small-prime part is unconditionally $O(\sqrt{n} \log n)$, so only the middle collision count can grow; but the exact avalanche count $K(n) = \#\{\sqrt{n} < p \leq n : p \mid G_n\}$ never exceeds 7 for $n \leq 15000$ (a counterexample needs $K(n) \gg n/\log n$), and the middle-prime multiplicity is identically 1 outside the range $n \leq 6$, so the full-multiplicity form (T) shows no phantom. A targeted search of the one structured locus is equally negative: the non-ordinary primes $p \mid \gamma(p)$ of 8.4.a.a below 12010 are only $\{11, 3137\}$, and at their forced apex indices $d = (p-1)/2$ the ratio $\log G_d/d$ is unremarkable while the forced prime does not divide G_d ; the coincidence of Lemma-11 type lives on an intercept- $O(1)$ line and does not concentrate at a single n . This is empirical and heuristic support, not a proof of (NC_η) .

7. UNCONDITIONAL STRUCTURAL THEOREMS

7.1. The gap eliminant U_g . Fix integers $g \geq 1$ and $r \geq 0$. Let $u^{(r)}$ be the rational solution of the Apéry recurrence initialized by

$$u_r^{(r)} = 0, \quad u_{r+1}^{(r)} = 1.$$

For concision, write $u_j = u_j^{(r)}$ while r is fixed.

Lemma 12 (Eliminant identity). *Let $p \geq 7$ and $0 \leq r < r+g < p$. If $r \in \mathbb{Z}_p$, then $p \nmid b_{r+1}$ and*

$$p \mid b_{r+g} \iff p \mid u_{r+g}.$$

Here the last divisibility means that the p -integral rational number u_{r+g} reduces to zero in $\mathbb{F}_p$.

Proof. Lemma 1 gives $p \nmid b_{r+1}$. On the interval $[r, r+g]$ all denominators in the recurrence are p -units. The two solutions b_j and $b_{r+1}u_j$ have the same initial values at $j = r, r+1$, so uniqueness for the recurrence gives

$$b_{r+j} \equiv b_{r+1}u_{r+j} \pmod{p} \quad (0 \leq j \leq g).$$

The assertion follows because b_{r+1} is a unit. $\square$

Recall the continuants $N_g(x)$ and $Q_g(x)$ from (1.2). That formula, applied to the normalized solution above, gives

$$Q_g(r)u_{r+g} = N_g(r), \quad Q_g(r) = \prod_{j=2}^g (r+j)^3.$$

Theorem E (Gap-eliminant theorem). *For every $g \geq 1$ there is a fixed polynomial*

$$U_g(x) := N_g(x) \in \mathbb{Z}[x]$$

with the following properties:

- (1) $\deg U_g = 3(g-1)$;
- (2) *the coefficient height of U_g is $\exp(O(g \log g))$;*
- (3) $U_g(r) > 0$ for every integer $r \geq 0$;
- (4) *if $p \geq 7$ and $0 \leq r < r+g < p$, then*

$$r, r+g \in \mathbb{Z}_p \implies p \mid U_g(r).$$

For $g = 1$, this means $U_1 = 1$ and the last assertion is vacuous by Lemma 1.

Proof. The identity preceding the theorem proves that clearing the denominators of u_{r+g} gives the fixed integer polynomial $U_g = N_g$.

Let L_g be the leading coefficient of N_g . From the continuant recurrence,

$$L_1 = 1, \quad L_2 = 34, \quad L_{g+1} = 34L_g - L_{g-1}.$$

Inductively $L_g > L_{g-1} > 0$. Hence no top-degree cancellation occurs, and $\deg N_g = 3(g-1)$.

For the height, let $\|F\|_1$ denote the sum of the absolute values of the coefficients of F . Shifting the fixed cubic P and the sixth power in the continuant recurrence gives

$$\|N_{g+1}\|_1 \leq C(g+1)^6 \max(\|N_g\|_1, \|N_{g-1}\|_1).$$

Iteration yields $\log \|N_g\|_1 = O(\sum_{j \leq g} \log j) = O(g \log g)$, which proves the height bound.

For positivity, the Wronskian identity gives

$$u_{r+g} = \frac{a_{r+g}b_r - b_{r+g}a_r}{a_{r+1}b_r - b_{r+1}a_r}.$$

The denominator is $W_r > 0$, and the numerator is positive because the positive sequence b_j has strictly increasing ratios a_j/b_j , again by $W_j > 0$. Thus $u_{r+g} > 0$ and consequently $U_g(r) = Q_g(r)u_{r+g} > 0$.

Finally, if $r, r+g \in \mathbb{Z}_p$, Lemma 12 gives $u_{r+g} \equiv 0 \pmod{p}$. Every factor of $Q_g(r)$ is a p -unit, so $p \mid U_g(r)$. $\square$

Corollary 1 (Bounded-gap pairs). *Uniformly for $1 \leq G \leq X$,*

$$\# \left\{ (p, r, g) : \begin{array}{l} X < p \leq 2X, \quad 1 \leq g \leq G, \\ 0 \leq r < r+g < p, \quad r, r+g \in \mathbb{Z}_p \end{array} \right\} \ll G^2 X.$$

Proof. For $r < 2X$ and $g \leq G \leq X$, the degree and height bounds give

$$\log U_g(r) \ll g \log X.$$

Since $U_g(r) > 0$, it has $O(g)$ prime divisors exceeding X . Sum this bound over $r < 2X$ and $g \leq G$. $\square$

Remark 8 (Scope of root distribution). Theorem E embeds bounded-gap double zeros into roots of a fixed polynomial, which is the algebraic prerequisite for the Hooley–DFI circle. It does not by itself prove that the Apéry subset of those roots is equidistributed. In particular,

$$U_2(r) = (2r + 3)(17r^2 + 51r + 39)$$

has a linear factor. The theorem of Duke, Friedlander, and Iwaniec treats roots of an irreducible quadratic to prime moduli [17], while Hooley’s general theorem orders roots by all moduli [27]. Thus the stronger prime-axis equidistribution claim in the working notes is not an unconditional corollary and is not used here.

7.2. A cross-arc eliminant.

Lemma 13 (Cross-arc lemma). *Fix $A > 0$. Uniformly for $1 \leq a, m \leq L$ and $y, y + m \in \mathcal{J}_X$, one has, for sufficiently large X ,*

$$(7.1) \quad |S_{a,X}(y) \cap S_{a,X}(y + m)| \leq \omega_{>X}(U_m(y)) \leq 4m.$$

Proof. If p belongs to the intersection, then $r = y - ap$ and $r + m = y + m - ap$ both lie in Z_p , with $0 \leq r < r + m < p$. Theorem E gives

$$p \mid U_m(r), \quad U_m(r) \equiv U_m(y) \pmod{p}.$$

Thus every such p divides the fixed positive integer $U_m(y)$. The degree and coefficient-height estimates in Theorem E, together with $y \leq XL$ and $m \leq L$, give

$$\log U_m(y) \leq (3 + o(1))m \log X.$$

It therefore has at most $4m$ prime divisors exceeding X for all sufficiently large X . $\square$

7.3. Structured resonance. For $R, H < X/2$, let $Z_p^{\text{str}}(R, H)$ consist of those $r \in Z_p$ which either satisfy

$$(7.2) \quad \min(r, p - 1 - r) \leq R \quad \text{or} \quad |2r - (p - 1)| \leq 2R,$$

or have a second zero $s \in Z_p$ with $0 < |r - s| \leq H$. Put

$$K_{R,H}(x) = \#\{p \in (X, 2X] : x \bmod p \in Z_p^{\text{str}}(R, H)\}.$$

Theorem F (Structured-resonance theorem). *Let $0 \leq \delta_0 < 1$, $2X \leq x \leq Y \leq X^{2+\delta_0}$, and $1 \leq R, H \leq X/4$. Then*

$$(7.3) \quad K_{R,H}(x) \ll_{\delta_0} R + H^2 + 1.$$

Consequently, with $(t)_3 = t(t-1)(t-2)$,

$$(7.4) \quad A_3^{\text{str}}(Y, X; R, H) := \sum_{2X \leq x \leq Y} (K_{R,H}(x))_3 \ll_{\delta_0} Y(R + H^2 + 1)^3.$$

The contribution in which all three primes hit the exact center $r = (p-1)/2$ is zero for sufficiently large X .

Proof. The anchor conditions force a block prime to divide the nonzero integer

$$L_R(x) = \prod_{0 \leq t \leq R} (x-t)(x+1+t) \prod_{|t| \leq R} (2x+1-2t).$$

There are $O(R+1)$ linear factors, and $\log |L_R(x)| \ll_{\delta_0} (R+1) \log X$; hence only $O_{\delta_0}(R+1)$ prime divisors of $L_R(x)$ exceed X .

If $x \bmod p$ has a second zero at distance $h \leq H$, then, according to which zero is smaller, Theorem E makes p divide either $U_h(x)$ or $U_h(x-h)$. Thus every close-pair prime divides

$$C_H(x) = \prod_{1 \leq h \leq H} U_h(x)U_h(x-h),$$

which is positive because $x-h > 0$. The degree and height of the factors give

$$\log C_H(x) \ll_{\delta_0} H^2 \log X.$$

It has $O_{\delta_0}(H^2)$ prime divisors exceeding X . This proves (7.3); summing its cube proves (7.4).

Finally, an exact-center hit gives $p \mid 2x+1$. Three distinct block primes would give $pqs \mid 2x+1$, whereas $pqs > X^3 > 2X^{2+\delta_0} + 1 \geq 2x+1$ for sufficiently large X . This is impossible. $\square$

7.4. Two lenses on the zero set.

Remark 9 (The Apéry–Hasse polynomial). For a prime p , define

$$(7.5) \quad \Phi_p(T) = \sum_{0 \leq k < p} \binom{T}{k}^2 \binom{T+k}{k}^2 \in \mathbb{F}_p[T].$$

Then $\deg \Phi_p = 4(p-1)$ and

$$Z_p = \{r \in \{0, \dots, p-1\} : \Phi_p(r) = 0\}.$$

The classical identity

$$b_r = {}_4F_3 \left[\begin{matrix} -r, -r, r+1, r+1 \\ 1, 1, 1 \end{matrix}; 1 \right]$$

has finite-field and p -adic counterparts. In particular, for $1 \leq r \leq p-2$, Kalita–Chetry express b_r modulo p as a Greene ${}_4F_3$ character sum, while McCarthy’s notation writes the same balanced family as ${}_4G(r/(p-1), r/(p-1), 1-r/(p-1), 1-r/(p-1))_p$ [30, 25, 35]. These are sums over characters, not bounded products, so they supply a moment lens rather than a compressive cross-prime invariant.

At the central index $r = (p-1)/2$, Ahlgren–Ono and Kilbourn give the modular-form congruences with the weight-4, level-8 newform $\eta(2\tau)^4 \eta(4\tau)^4$ [1, 32]. The Apéry index is $(p-1)/2$, not $p-1$; indeed $b_{p-1} \equiv 1 \pmod{p}$.

Remark 10 (Projective Frobenius lens). Let $m = (p - 1)/2$ and let $\gamma(p)$ be the p th coefficient of $\eta(2\tau)^4\eta(4\tau)^4$. Reducing the recurrence around m produces universal rational continuant coefficients A_j, B_j , on every local interval where the recurrence denominators are p -units, such that

$$(7.6) \quad \begin{aligned} b_{m+j} &\equiv A_j\gamma(p) + B_jb_{m-1} \pmod{p}, \\ (A_0, B_0) &= (1, 0), \quad (A_1, B_1) = (0, 1). \end{aligned}$$

The Wronskian prevents the two seeds from vanishing simultaneously. Thus the root condition can be read projectively as

$$(7.7) \quad [A_j : B_j] = u_p, \quad u_p = [-b_{m-1} : \gamma(p)] \in \mathbb{P}^1(\mathbb{F}_p).$$

This is complementary to (7.5): a fixed continuant curve meets a moving projective Frobenius datum. No cross-prime anti-concentration theorem for u_p is asserted here. Newton–Thorne symmetric-power functoriality controls the archimedean Sato–Tate distribution, not the self-characteristic condition $p \mid \gamma(p)$ [38].

8. DIFFICULTY CERTIFICATION

8.1. Phantom locality.

Remark 11 (Phantom-locality doctrine). Every currently available holonomic divisibility constraint in this problem is local at one prime: the Lucas and Dwork congruences, reflection, the Wronskian and Casoratian, the U_g and cross-arc certificates, coefficient heights, and the polynomial Φ_p . None supplies a map from zero positions at different primes to a common object with substantially fewer fibers. They therefore do not exclude the possibility that, at many primes, one locally admissible zero lies at $z_p \equiv y_0 - ap \pmod{p}$. Reflection adds a partner locally and the eliminants constrain local pairs, but neither operation couples the chosen positions across different primes. Such an abstract arc phantom would create one large exact fiber.

This is a limitation of the banked invariant list, not a construction of alternative roots of the actual polynomial Φ_p . In particular, it does not prove logical independence from all possible Apéry identities. It identifies the missing kind of input: a cross-prime compressive invariant. The multiplicative-order fibration in Lemma 14 below is the model of such an invariant.

The usual single-value eliminant is b_y itself, whose exponential height reproduces the trivial divisor bound. Hermite–Padé denominator bounds, the Casoratian, and fixed-prime hypergeometric identities do not alter this cross-prime obstruction. The available Apéry factorization and congruence results [12, 16, 42] provide strong local information but no such fibration.

8.2. A fibration calibration.

Lemma 14 (Mersenne fibration). *Fix $A > 0$. Uniformly for $1 \leq a \leq L$ and $y \in \mathcal{J}_X$, the admissible Mersenne arc satisfies*

$$(8.1) \quad \#\{p \in (X, 2X] : 0 \leq y - ap < p, p \mid 2^{y-ap} - 1\} \ll_A X^{1/2+o(1)}.$$

Proof. Let $d_p = \text{ord}_p(2)$. The divisibility condition is $d_p \mid y - ap$. Since $d_p \mid p - 1$, it follows that

$$(8.2) \quad d_p \mid y - a.$$

Here $0 < y - a \ll_A X(\log X)^A$, so $\tau(y - a) = X^{o(1)}$ uniformly.

For a divisor $d \leq \sqrt{X}$ of $y - a$, every prime with $d_p = d$ divides $2^d - 1$. The number of its prime divisors exceeding X is $O(d/\log X)$. Summing over the possible d costs $X^{1/2+o(1)}$. If $d > \sqrt{X}$, then $p \equiv 1 \pmod{d}$, so the interval $(X, 2X]$ contains $O(X/d + 1) = O(\sqrt{X})$ candidates for each such divisor. Another use of $\tau(y - a) = X^{o(1)}$ proves (8.1). $\square$

The restriction to the in-range window is essential: with unrestricted y , the integer $y - a$ could be a common multiple of all relevant orders. The lemma exhibits exactly what Apéry lacks. Mersenne zero positions fiber through the divisor lattice of one small integer $y - a$; Apéry zero positions are not known to fiber through a subgroup or any other compressive cross-prime object.

8.3. The horizontal Fourier barrier. Put

$$(8.3) \quad W_X(h) = \sum_{X < p \leq 2X} \sum_{m \in \mathbb{Z}_p} e\left(\frac{hm}{p}\right), \quad \mathcal{S}_X = \{p \in (X, 2X] : p \mid \gamma(p)\},$$

where $\gamma(p)$ is the coefficient in $\eta(2\tau)^4 \eta(4\tau)^4$. Ahlgren–Ono identify

$$(8.4) \quad p \in \mathcal{S}_X \iff (p - 1)/2 \in \mathbb{Z}_p.$$

Theorem G (Barrier R9.2: Lang–Trotter hardness). *Let $2 \leq H \leq X$. There is an absolute $c > 0$ such that*

$$(8.5) \quad cH|\mathcal{S}_X| \leq \mathcal{M}_X + 2 \sum_{1 \leq h < H} |W_X(h)|.$$

Consequently, if for some fixed $\delta > 0$ one has, with $H = \lfloor X/(4L) \rfloor$,

$$(8.6) \quad \max_{1 \leq h < H} |W_X(h)| \ll X^{1-\delta},$$

then $|\mathcal{S}_X| \ll X^{1-\delta'}$ for some $\delta' > 0$. The same conclusion follows from

$$(8.7) \quad \sum_{1 \leq h < H} |W_X(h)|^2 \ll HX^{2-2\delta}.$$

In particular, either horizontal estimate implies that the primes p for which $p \mid \gamma(p)$ have density zero.

Proof. For the Fejér kernel

$$F_H(t) = \frac{1}{H} \left| \sum_{j=0}^{H-1} e(jt) \right|^2 = \sum_{|h| < H} \left(1 - \frac{|h|}{H} \right) e(ht),$$

positivity gives

$$\begin{aligned} 0 &\leq \sum_{|h| < H} \left(1 - \frac{|h|}{H} \right) e(-h/2) W_X(h) \\ &= \sum_{X < p \leq 2X} \sum_{m \in \mathbb{Z}_p} F_H \left(\frac{m}{p} - \frac{1}{2} \right). \end{aligned}$$

If $m = (p-1)/2$ and $H \leq X < p$, then $F_H(-1/(2p)) \geq cH$. Keeping these nonnegative terms and bounding the nonzero Fourier modes in absolute value proves (8.5).

Lemma 5 gives $\mathcal{M}_X \ll X^{5/3+o(1)}$. Divide (8.5) by $H \asymp X/L$ and use (8.6); the two terms are $X^{2/3+o(1)}$ and $X^{1-\delta}$, up to powers of $\log X$. This gives a fixed power saving. Under (8.7), use Cauchy's inequality before dividing by H . $\square$

The form 8.4.a.a is non-CM of weight 4. No theorem currently gives even $o(\pi(X))$ non-ordinary primes for this fixed form. Deligne's bound allows $\gamma(p) = cp$ with $|c| \ll \sqrt{p}$ and does not force vanishing, while Newton–Thorne controls an archimedean Sato–Tate distribution rather than divisibility by the varying prime. Thus Theorem G is an implication to a recognized open Lang–Trotter-type problem, not a claim that (8.6) is logically unprovable.

Remark 12 (Barrier R9.1: bandwidth insufficiency). For a planted arc $m_p \equiv y_0 \pmod{p}$, the summand in $W_X(h)$ is $e(hy_0/p)$. Reciprocal-phase cancellation can make the high-frequency range look equidistributed while all planted primes still collide at the exact integer y_0 . The working-note R9.1 proposes correcting the low frequencies by a small perturbation among the remaining primes, thereby producing an abstract full-bandwidth phantom. The notes do not supply a positivity- and integrality-preserving realization as actual root sets of the fixed polynomials Φ_p , so this stronger construction is recorded as a barrier model rather than a theorem.

Its robust conclusion is categorical: Fourier bounds for the normalized points m/p control intervals of positive width. The condition $ap + m = aq + w$ is an exact equality at scale zero. Without an input tied to that equality, horizontal equidistribution is neither the terminal statement nor a replacement for (NC_η) .

8.4. The scale-zero core. Identity (6.10) is the corrected detector. Opening $N_{\text{coll},a}$ by the Chinese remainder theorem and applying additive orthogonality produces a reciprocity-twisted quadrilinear form with moduli pq . Away from the locus on which the two reciprocal phases recombine, this has the formal shape treated by the bilinear and trilinear Kloosterman-fraction

estimates of Duke–Friedlander–Iwaniec and Bettin–Chandee [18, 7]. No estimate for the Apéry-weighted form is proved here; attaching those tools to the non-degenerate periphery remains a prospective step.

Remark 13 (Degeneracy certificate). On the recombination locus, an arc phantom makes both phases combine to a constant depending on the common exact value. The mass on that locus is precisely $N_{\text{coll},a}$, so an equidistribution estimate for the complementary locus leaves the terminal conjecture unchanged. Rationally aligned central clusters do not import the weight-4 ordinarity obstruction into $N_{\text{coll},a}$: after clearing the fixed denominator, the collision congruence is a multiple of pq whose absolute value is smaller than pq , and hence is an integer equality. This is why the horizontal Fourier route asks for strictly more than the exact-collision reduction needs.

Remark 14 (Current technology boundary). The nearest exact-divisibility tools exploit multiplicative structure: large prime factors in Lucas and Lehmer sequences [48], or Subspace-Theorem bounds for greatest common divisors of S -unit values [15]. The available results on Apéry numbers are one-sided valuation or Lucas-congruence statements [33, 16, 42]; they do not give an upper bound for $\omega_{(X,2X]}(b_y)$. No standard GRH-type hypothesis presently formulates, let alone implies, (Ω_η) .

The two live directions are therefore: construct a cross-prime compressive invariant for the vanishing positions of the mod- p Picard–Fuchs solutions, or develop an exact-collision tool for holonomic values. The non-degenerate Kloosterman-fraction periphery [18, 7] is the natural first test case; matching the Apéry weights to the hypotheses of those methods, so as to obtain a theorem-level bound on the provable periphery of (Ω) , remains to be carried out.

8.5. The six-fold barrier map. The preceding barriers admit a common organization. The word “complete” below is relative to the audited mechanisms: five ensemble methods and one single-index adelic method. A barrier certificate says that the stated input class cannot distinguish the terminal event; it is not a logical independence result for arbitrary future mathematics.

8.5.1. Sieve complexity and the integrality criterion. For a family of p -local relations $\mathcal{R}_p$, define its *sieve complexity* $c_X(\mathcal{R})$ to be the least number of global integral relations whose reductions cover $\mathcal{R}_p$ for all $p \in (X, 2X]$. Only the order of magnitude of this invariant will be used. The calibrating examples are

(8.8)

relation	c_X	status of the corresponding cover
weight-3 ordinarity	$O(1)$	provable
weight-4 form 8.4.a.a	$X^{1/2}$	open
Mersenne divisibility	$X^{o(1)}$	provable
Apéry divisibility at index n	$e^{\Theta(n)}$	outside the subexponential range.

For the first row, Deligne's interval $|a_p| \leq 2p$ leaves the five integer classes $a_p = cp$. For weight 4, the interval $|\gamma(p)| \leq 2p^{3/2}$ leaves $\asymp \sqrt{X}$ such classes. In the Mersenne model, the classes are the divisors of $y - a$, as in (8.2). For Apéry numbers, writing a divisibility relation as a global equality introduces a cofactor of exponential range.

Proposition 3 (Integrality criterion). *A fixed-quotient Galois or sieve argument can test a relation uniformly across varying primes only after that relation has been expressed by global integral classes. A condition defined only in $\mathbb{F}_p$ has no reduction in a fixed auxiliary quotient. Consequently, such a method applied to $n \bmod p \in Z_p$ must first pay the covering cost c_X in (8.8).*

Proof sketch. A global equality over $\mathbb{Z}$ may be reduced modulo every fixed auxiliary prime and is therefore visible to a fixed-quotient sieve. By contrast, the residue $z_p \in \mathbb{F}_p$ belongs to a different fiber for each p ; there is no canonical image of the equality $n \equiv z_p \pmod{p}$ in a fixed quotient. Replacing it by global equalities amounts precisely to enumerating its integral lifts. The four counts following the table give the asserted calibration. Thus the criterion does not say that large image never detects equality; it separates globally integral equalities, which it can detect, from self-characteristic p -local equalities, which require a cover. $\square$

8.5.2. The five ensemble barriers.

Proposition 4 (Barrier R9: analytic closure). *The horizontal Fourier route is closed in two complementary senses. First, the estimates (8.6) or (8.7) imply a density-zero theorem for the weight-4 nonordinary primes through Theorem G, an open Lang–Trotter-type endpoint. Second, the bandwidth model in Remark 12 can preserve an exact planted collision while matching the available positive-width distribution data. In particular, the exact-collision count (NC_η) is a scale-zero statistic not determined by horizontal equidistribution.*

Proof sketch. The first assertion is the Fejér-positivity argument (8.5). The second is the reciprocal-phase calculation in Remark 12. That calculation is a barrier model, not a realization by the actual roots of Φ_p ; the robust conclusion is only that interval discrepancy and exact equality belong to different categories of information. $\square$

Proposition 5 (Barrier R10a: collision decoupling). *For distinct $p, q \in (X, 2X]$, an exact collision factors into*

$$(8.9) \quad [z \in Z_p] \quad \times \quad [w \in Z_q] \quad \times \quad [ap + z = aq + w \in \mathcal{J}_X].$$

The first two factors are separate self-characteristic conditions, while the last factor is a Frobenius-free CRT window condition. There is no joint algebraic relation between the projective data u_p and u_q in (8.9). Accordingly, joint large-image, Mumford–Tate, and fixed-quotient Chebotarev information does not by itself make the collision rare.

Proof sketch. For fixed p and q , CRT makes the two residue conditions independent; the only coupling is the requirement that their common lift lie in the short archimedean window. Each local factor lives in its own characteristic and $b_{(p-3)/2}$ is not a common Weil-number datum. Even a square-root Chebotarev error for a pair has size $(pq)^{1/2} \asymp X$; summing such errors over prime pairs is larger than the target. Finally, the abstract coherent zero-section from Remark 11 is compatible with maximal fixed-quotient and Sato–Tate data while failing (SA). This is the Galois-phantom certificate. As before, it is a certificate for the stated hypothesis class, not a second system of roots for Φ_p . $\square$

Put

$$\lambda = \log(17 + 12\sqrt{2}) = 4\log(1 + \sqrt{2}).$$

Proposition 6 (Barrier R10b: the $3n$ determinant cap). *If an archimedean one-form argument uses an irrationality exponent $\mu \geq 2$ for $\zeta(3)$, then its resulting coefficient is bounded by the template*

$$(8.10) \quad \log G_n \leq n \frac{(\mu - 1)(3 + \lambda) - (\lambda - 3)}{\mu} + o(n).$$

At the Rhin–Viola value $\mu = 5.51389\dots$ this is $5.24673\dots n$, while the best conceivable limit $\mu \rightarrow 2$ is exactly $3n \sim \log d_n$. Thus neither this one-form method nor its Hermite–Padé determinant localization can yield $o(n)$ without new p -local input.

Proof sketch. The right-hand side of (8.10) equals $3 + \lambda - 2\lambda/\mu$, which is the corrected Rhin–Viola calculation recorded above. Sending μ to the Roth threshold 2 leaves 3. In a multi-form determinant the Apéry row still carries the denominator d_n ; the Casoratian has only polynomial height and prime support below n . Hence localization reproduces, rather than removes, the same $3n$ budget. This is the point at which general Subspace-Theorem and multiplicative-group technology [43, 20, 11] returns the problem to the missing local support law. $\square$

Consider the projective skew product over the shift $t \mapsto t+1$ with transfer matrix

$$(8.11) \quad M(t) = \begin{pmatrix} P(t) & -t^3 \\ (t+1)^3 & 0 \end{pmatrix}.$$

Scalar normalization is irrelevant projectively.

Proposition 7 (Barrier R11: torus versus SL_2). *Let G be the difference-Galois group over $\overline{\mathbb{Q}}(t)$ of the rank-2 Apéry difference module defined by (8.11).*

(NI-1) *One has $G \supseteq \mathrm{SL}_2$.*

(NI-2) *For each fixed degree bound D , a nonconstant invariant $F \in \mathbb{F}_p(x, t)$ of degree at most D for infinitely many p would lift to characteristic zero. Hence, outside finitely many primes, the mod- p cocycle has no*

uniformly bounded-complexity first integral beyond the determinant line, represented here by the Casoratian.

The Mersenne comparison has toric difference-Galois group and a full character lattice of invariants. Thus the torus/ SL_2 dichotomy identifies large monodromy with the failure of a compressive first integral.

Proof sketch. Birkhoff asymptotics at infinity leave characteristic factors $\alpha^{\pm t} t^{-3/2}(1 + \dots)$, with $\alpha = 17 + 12\sqrt{2}$. Substitution in the hyperexponential possibilities excludes a rank-1 solution; the half-integral exponent is the terminal certificate in this calculation. A dihedral image would give a hyperexponential solution of the symmetric-square recurrence, whose three candidate exponential ratios are $\alpha^2, 1, \alpha^{-2}$. The terminating Hyper computation excludes them, in the Hendriks–Singer classification [26]. Irreducibility, primitivity, and infinitude then give (NI-1).

For (NI-2), the coefficients of an invariant of degree at most D satisfy a finite linear system over $\mathbb{Z}[t]$. Its rank can drop modulo only the finitely many primes dividing the relevant nonzero minors. Invariants for infinitely many primes would therefore lift, contradicting (NI-1). Algebraic entropy alone is not the relevant test: the base shift makes the skew product have zero algebraic entropy while its invariant ring remains minimal [6]. The Lucas–Gessel factorization at $t \equiv -1, 0 \pmod{p}$ is instead an arithmetic singularity-confinement event, in the sense of almost-good reduction [29]; it supplies the already-used digit law, not a cross-prime first integral. $\square$

Proposition 8 (Barrier R12: star collapse and the three habitats). *Let*

$$\mathcal{T}_X = \{(a, p, z) : 1 \leq a \leq L, p \in (X, 2X], z \in \mathbb{Z}_p, ap + z \in \mathcal{J}_X\}$$

and let $r_X(y)$ be the number of representations $y = ap + z$ from $\mathcal{T}_X$. Then the following certificates hold.

- (1) *If $\mathcal{A}_X = \{ap : 1 \leq a \leq L, p \in (X, 2X]\}$ and $M = |\mathcal{A}_X|$, then*

$$(8.12) \quad E_+(\mathcal{A}_X) \gg \frac{M^3}{\log X}.$$

Thus the prime-slope set already has nearly maximal additive energy.

- (2) *The collision graph on $\mathcal{T}_X$ is a disjoint union of cliques indexed by y , and*

$$(8.13) \quad N_*(X) := \sum_y r_X(y)(r_X(y) - 1), \quad \sum_y r_X(y) = |\mathcal{T}_X|.$$

Under (C1), a power saving for $N_(X)$ is equivalent, after renaming exponents, to a power saving for $\max_y r_X(y)$, hence to the power-saving forms of (SA) and (Ω).*

- (3) *The coherent section $z_p \equiv y_0 \pmod{p}$ has the same fiber-size, prime-slope, and normalized-distribution inputs as the abstract phantom, but it creates one fiber of large multiplicity. In the line model $v = pu + z_p$, writing $z_p = y_0 - b_p p$ with $b_p = \lfloor y_0/p \rfloor$ makes every line*

meet the horizontal fiber $v = y_0$ at an allowed integer $u = b_p$. After the section-dependent trivialization $c = u - b_p$, this is the projected pencil with apex $(0, y_0)$.

Consequently, additive combinatorics has only three relevant habitats: a fixed auxiliary quotient, closed by Proposition 3; the individual fibers $\mathbb{F}_p$, where singleton-scale sets have no internal structure; and global $\mathbb{Z}$, where (8.12) and (8.13) return the argument to the supremum cap. Within these hypotheses, sum-product and incidence methods are phantom-transparent.

Proof sketch. Distinct products ap are unique for $a \leq L = o(X)$, so $M \asymp LX/\log X$, while $\mathcal{A}_X + \mathcal{A}_X$ lies in an interval of length $O(LX)$. Cauchy's inequality gives $E_+(\mathcal{A}_X) \geq M^4/|\mathcal{A}_X + \mathcal{A}_X|$, proving (8.12).

Two progressions with distinct moduli p, q can meet at most once in the window: two meetings would make the differences of their slope parameters multiples of q and p , larger than L . The same-prime representation is unique. This proves the clique decomposition. If $R = \max_y r_X(y)$ and $T = \sum_y r_X(y)$, then

$$R(R-1) \leq N_*(X) \leq RT.$$

Under (C1), $T \leq L\mathcal{M}_X = X^{1+o(1)}$, giving the claimed equivalence after exponent renaming. The third assertion is the direct substitution $y_0 = b_p p + z_p$. It is a certificate for hypotheses built from fiber statistics, mass, prime distribution, and equidistribution; it does not assert that the coherent section is the actual Apéry zero-section. Bilinear and incidence estimates can still treat a nondegenerate periphery [23, 9, 18, 7], but the concentrated fiber is exactly the quantity in (Ω) . $\square$

8.5.3. The single-index adelic barrier. Let $D_n = \text{lcm}(1, \dots, n)$, so that $d_n = D_n^3$, and put

$$(8.14) \quad \mathcal{B}_n = D_n^3 b_n, \quad \mathcal{A}_n = D_n^3 a_n, \quad \ell_n = b_n \zeta(3) - a_n.$$

The standard Apéry remainders ℓ_n are positive and satisfy $|\ell_n| \asymp e^{-\lambda n} n^{-3/2}$.

Lemma 15 (Casoratian floor). *For every $n \geq 0$,*

$$(8.15) \quad |\ell_n| \geq \frac{6}{(n+1)^3 b_{n+1}} = e^{-\lambda n - O(\log n)}.$$

Proof. The Casoratian identity gives

$$b_{n+1} \ell_n - b_n \ell_{n+1} = a_{n+1} b_n - a_n b_{n+1} = \frac{6}{(n+1)^3}.$$

Both remainders have the same sign, so $b_{n+1} |\ell_n| \geq 6/(n+1)^3$. The known growth $b_{n+1} = e^{\lambda n + O(\log n)}$ proves the last equality. $\square$

Proposition 9 (Barrier R13a: the critical adelic ledger). *For $x_n = (\mathcal{B}_n : \mathcal{A}_n) \in \mathbb{P}^1(\mathbb{Q})$,*

$$h(x_n) = (\lambda + 3)n + O(\log n).$$

The best-form bookkeeping from the Apéry lattice has total local smallness

$$(3\lambda + 3)n + O(\log n),$$

whereas the rank-2 Roth–Subspace threshold is $2h(x_n) = 2(\lambda + 3)n + O(\log n)$. The surplus is therefore exactly

$$(8.16) \quad (\lambda - 3)n + O(\log n),$$

the same margin used in Apéry’s irrationality argument. It is not a second budget that can also pay for locating the primes dividing b_n .

Proof sketch. The height is the sum of the exponential growth λn of the Apéry solution and the denominator-clearing tax $3 \log D_n = 3n + o(n)$. Tallying the best local form at infinity and at the finite places gives $(3\lambda + 3)n$; subtracting twice the height gives (8.16). The only form carrying this positive surplus at infinity has coefficient $\zeta(3)$ and is therefore not an algebraic-coefficient form to which the Subspace Theorem applies. Removing it leaves the ledger exactly critical. Lemma 15 also shows that the determinant and archimedean channels are two readings of one identity and cannot be added. $\square$

Proposition 10 (Barrier R13b: the remaining adelic gates). *The single- n , many-place route meets the following four additional certificates.*

- (1) Partition blindness. *The logarithm of the contribution of primes in $(X, 2X]$ is an internal partition of the fixed mass $\log b_n$. Heights weigh that divisor but do not locate its support. Every algebraic arithmetization of (8.14) pays the D_n^3 tax, which is divisible at all window primes and hence has no localizing content.*
- (2) Unit-points versus approximation-vectors. *The Subspace-Theorem amplification for S -unit points uses coordinate mass and form smallness on disjoint place sets. Here the coordinate b_n itself vanishes at each moving prime, so the input is the unknown middle-prime mass. A manufactured second integer can help only when the residues share a fixed lift z' ; then every window prime divides $n - z'$, giving only polylogarithmically many primes unless $n = z'$. The latter is precisely the coherent apex excluded by (Ω) .*
- (3) Wrong sign and wrong disc. *The André–Chudnovsky denominator theory contributes the same D_n^3 tax, not a denominator gain. Available p -adic G -function transcendence works inside a strict p -adic disc, whereas the Dwork evaluations defining Z_p occur at Teichmüller boundary points of absolute value 1 [2, 13, 19]. Uniformly reaching that boundary would already require the missing unit-root information.*

- (4) Skeleton frontier and radical transparency. *The current frontier is summarized by*

(8.17)

<i>sequence</i>	<i>apparition skeleton</i>	<i>status</i>
$2^n - 1$	$\text{ord}_p(2) \mid n$	$X^{o(1)}$ elementary
<i>two-torus S-units</i>	<i>finite-rank lattice</i>	<i>unconditional</i>
<i>elliptic divisibility</i>	<i>elliptic apparition law</i>	<i>Vojta-conditional</i>
<i>Hecke coefficients</i>	<i>multiplicative, ensemble-level</i>	<i>open</i>
<i>Apéry numbers</i>	<i>no known apparition law</i>	<i>beyond this frontier.</i>

Moreover, an *abc* argument based on radical versus height sees the excess $(v_p - 1) \log p$ but is transparent to the valuation-one primes that make up the radical target (Ω) . The valuation-one phantom therefore passes this input as well.

Proof sketch. The first item is invariance of projective height under repartitioning the finite-place contribution. The second contrasts the finite-rank-unit mechanism behind the Corvaja–Zannier gcd theorem [11] with the self-referential coordinate b_n ; the fixed-lift assertion follows because the product of distinct window primes dividing the nonzero integer $n - z'$ is bounded by its size. The third item is the location of the known G -function estimates relative to the Teichmüller boundary. The fourth records the progressively weaker multiplicative skeleton; the conditional abelian-variety frontier is represented by Silverman’s Vojta analysis [46]. Finally, valuation-one primes do not enlarge the difference between the logarithm of an integer and the logarithm of its radical, which proves the stated *abc*-transparency certificate. $\square$

8.5.4. Composition and the missing comparison.

Proposition 11 (Two unifying principles). *The six barriers compose without circularity.*

- (1) Phantom-transparency: *the five ensemble families use statistics unchanged when the true zero-section is exchanged for the coherent projected-pencil section, whereas (SA) is changed.*
- (2) Critical adelic ledger: *the single- n arithmetic family does not use that exchangeability; it stops because its algebraic-form surplus is exactly the already-spent irrationality margin (8.16), and because heights cannot locate a divisor’s support.*

Thus the first five barriers and the sixth adelic barrier are independent certificates for the two proposed exits.

Proof sketch. The first five barrier propositions inspect ensemble or fixed-quotient data. Proposition 10 instead fixes one n and audits the product-formula ledger across its places. Its obstruction does not assume the ensemble phantom, while the phantom remains compatible with its radical and

height data. No premise of either certificate is therefore borrowed from the desired conclusion of the other. $\square$

Remark 15 (Scope of the six-fold map). Current multi-place tools factor through one or more of three kinds of input: heights, separate per-place data, and ensemble statistics. Partition blindness, the boundary-disc obstruction, and phantom-transparency respectively block those three inputs. In this precise methodological sense the audit leaves no seventh current family. This is an optimality statement for the reduction and the audited technology, not a theorem that an unknown kind of comparison cannot exist.

Remark 16 (The missing product formula for periods). The remaining target may be named a *product formula for periods*: a place-mixing inequality for the Apéry motive at one integer index n , comparing its Betti/de Rham realization, represented by the archimedean period remainder ℓ_n , with its crystalline realizations at every $p \in (X, 2X]$, represented by digit evaluations of the Frobenius crystal at Teichmüller boundary points. Such an inequality would have to imply a power saving in (Ω) without factoring through an algebraic avatar that pays the universal D_n^3 tax. This is a description of the required new conjectural input, not a theorem supplied here. Neither the standard GRH-type information audited above nor *abc* provides this place-mixing comparison.

8.5.5. *The seventh barrier: constructing the comparison.* Write (B) for the terminal exact-collision target (NC_η) , or, under (C1) and after renaming positive exponents, for either of its equivalent forms (SA) and (Ω) . The preceding discussion characterizes the desired output of a place-mixing construction. Indeed, if a nonzero integer C_n is divisible by every prime in $(X, 2X]$ which divides b_n , then

$$(8.18) \quad \omega_{(X, 2X]}(b_n) \leq \frac{\log |C_n|}{\log X}.$$

Thus an avatar-integer proof of (Ω) would have to produce a window-supported C_n with $\log |C_n| \ll X^{1-\eta} \log X$. The ordinary holonomic avatars have logarithmic height comparable with n , which is $X(\log X)^{O(1)}$ in the in-range window. The missing saving is a genuine power of X , not a logarithmic refinement of the height ledger.

Proposition 12 (Barrier R14: the construction gate). *The three proposed constructions of the comparison in Remark 16 meet the following five certificates. Each assertion is relative to the indicated construction formalism.*

- (1) *Functoriality and the invariant ring.* A bounded complexity scalar constructed functorially from the fixed rank-2 difference module lies in its invariant calculus. Parts (NI-1) and (NI-2) of Proposition 7 show that this calculus factors through the determinant, or Casoratian, line. Its logarithmic height is $O(\log n)$, but its prime support is confined to the recurrence singularities $n \equiv 0, -1 \pmod{p}$

and is therefore window-blind. In the Birkhoff-weight calculus, every nonzero weight is on the exponential scale e^{cn} . The resulting height spectrum is gapped: it contains no window-supported avatar in the intermediate slot required by (8.18). Multiplying the available per-prime continuant certificates across p simply restores the full height ledger.

- (2) No cross-prime interpolation. The digit value $b_r \bmod p$ is represented by a finite-field hypergeometric datum whose parameter $r/(p-1)$ moves with p along $r = n \bmod p$ [30]. These formulas therefore do not place the moving data in the fixed-object interpolation spaces used by motivic or p -adic L -function constructions. The fixed-parameter channels are sparse for an elementary reason: if

$$\frac{r}{p-1} = \frac{a}{k} \quad \text{and} \quad r \equiv n \pmod{p}$$

with a/k fixed, then $p \mid kn + a$. For fixed a/k these channels contain only $O(\log n)$ prime divisors, and the central channel $a/k = 1/2$ is exactly $p \mid 2n + 1$, where the Ahlgren–Ono modular-form congruence applies [1]. Existing L -value and regulator constructions for a fixed object consequently see either these torsion-digit channels or global heights; neither output locates the generic moving digit.

- (3) Elliptic-divisibility calibration. Even an exact place-by-place height formula is not a locating theorem. Elliptic divisibility sequences possess such a formula and, in addition, a fixed-quotient apparition law, yet their effective window problem still passes through growing-level Chebotarev estimates. The precise calibration is recorded in Remark 17 below.
- (4) Boundary and non-overconvergence. Fontaine–Laffaille and integral p -adic Hodge comparison theorems provide functorial filtration and lattice divisibilities [22, 8]. Their uniform part is independent of the digit and hence reproduces the universal D_n^3 tax. At Teichmüller digit points, the available sequence-specific output remains the Beukers–Vlasenko vertical tower

$$b_{mp^k} \equiv b_{mp^{k-1}} \pmod{p^{3k}} \quad (p \geq 5),$$

in its applicable range [14, 5]. This compares p -power indices at one prime; it neither couples two primes nor locates $n \bmod p$ inside Z_p . Integral refinement does not lower the intrinsic linear-in- p complexity of the digit polynomial Φ_p or supply the missing uniform continuation from a strict p -adic disc to all Teichmüller boundary points.

- (5) Direction of place mixing. *The audited local-to-global theorems run from strong information at many primes to a rigid global conclusion: Fontaine’s nonexistence theorem, the Grothendieck–Katz p -curvature paradigm, and arithmetic algebraization are standard examples [21, 31, 2, 10, 13]. Target (B) asks for the reverse direction, from soft archimedean information to a cap on a conspiracy of self-characteristic local events. The Pila–Zannier strategy likewise requires a bounded-complexity family of special loci [39]; the available defining polynomials Φ_p have degree $4(p-1)$, and no bounded-complexity replacement has been supplied. Consequently this strategy, without a new uniformity theorem, re-enters the same height ledger rather than producing (8.18). In this audited sense there is no eighth current family beyond the construction gate.*

Proof sketch. The first item is Proposition 7 applied to functorial scalars, together with the two Birkhoff weights used in its proof. The invariant line gives only the Casoratian factors; a non-invariant covariant has nonzero difference weight. The second item uses the displayed congruence $kn \equiv -a \pmod{p}$; counting prime divisors of the fixed nonzero integer $kn+a$ gives the stated logarithmic bound. The third item is Remark 17. In the fourth item, filtration divisibility is uniform in the point, whereas the displayed Dwork tower is vertical at a fixed prime. The final item compares the directions and the complexity hypotheses of the cited theorems. None of these arguments claims logical independence from an unknown construction; they certify the limits of the stated mechanisms. $\square$

Remark 17 (EDS calibration: a product formula that still does not locate). Let $E/\mathbb{Q}$ be an elliptic curve, let $P \in E(\mathbb{Q})$ be nontorsion, and write, on a minimal model,

$$x(mP) = \frac{A_m}{B_m^2}, \quad (A_m, B_m) = 1.$$

At every prime of good reduction, the finite local Néron height has the exact denominator term

$$\lambda_p(mP) = v_p(B_m) \log p.$$

After the standard corrections at the finitely many bad places, the local heights satisfy the exact global decomposition

$$(8.19) \quad \sum_v \lambda_v(mP) = \hat{h}(mP) = m^2 \hat{h}(P).$$

Moreover, if r_p is the order of the reduction of P in $E(\mathbb{F}_p)$, then

$$p \mid B_m \iff r_p \mid m, \quad r_p \mid \#E(\mathbb{F}_p).$$

These apparition conditions are visible in finite division and Kummer quotients [45]. Nevertheless, the standard route to a one-sequence window cap must count Frobenius conditions in levels growing with the window; estimates of the needed uniform strength are GRH-grade, while unconditional

effective Chebotarev is too weak for that route. For two independent divisibility sequences, Silverman’s strong gcd cap used in this calibration is conditional on Vojta’s conjecture [46]. Thus the EDS model possesses both an exact place-by-place ledger and a fixed-quotient apparition law, but these do not by themselves give the required unconditional locating estimate. The product formula is the necessary *format* of the proposed comparison, not its locating content.

Conjecture 8 (Quotient-free locating for the Apéry loci). *The terminal bound (NC_η) holds for the self-characteristic loci $n \bmod p \in Z_p$. Equivalently under (C1), after renaming the positive exponent, either (SA_η) or (Ω_η) holds.*

This is the same numerical assertion as Conjecture 7, not an additional hypothesis. The new name records the mechanism that is missing: an effective apparition/equidistribution theorem for entropy-positive p -local loci which does not first map the events to a fixed auxiliary quotient. No theorem in the cited p -adic Hodge, GRH–Chebotarev, *abc*, Vojta, or unlikely-intersection frameworks supplies this quotient-free statement.

Proposition 13 (The seven-fold map and three unifying principles). *Within the scope stated in the barrier propositions, the analytic, algebraic geometry, Diophantine, dynamical, additive combinatorics, adelic, and motivic construction barriers compose without circularity. Their obstructions are organized by three principles:*

- (1) Phantom-transparency for the five ensemble families;
- (2) the critical adelic ledger for algebraic single-index avatars; and
- (3) quotient-free locating: *every locating mechanism in the calibrated siblings factors through a fixed quotient, while the Apéry digit event has no bounded-complexity quotient in the audited invariant calculus.*

The missing statement after all seven barriers is Conjecture 8, not merely the existence of a period comparison.

Proof sketch. Proposition 11 composes the first six barriers. Proposition 12 does not use a replacement zero-section or spend a second copy of the adelic surplus: it audits the proposed construction of the comparison itself. The EDS calibration then separates a place-by-place ledger from the additional act of locating its support. Hence the seventh certificate adds the third principle without borrowing the desired collision bound. $\square$

Remark 18 (Relation to the six-fold scope statement). The phrase “no seventh current family” above referred to output formats already covered by heights, separate per-place data, and ensemble statistics. Proposition 12 instead treats the previously unaudited attempt to *construct* the proposed comparison and records it as the seventh barrier. Its direction-of-place-mixing item preempts an eighth family assembled from the presently cited tools. Both scope claims remain methodological, not logical-independence theorems.

Remark 19 (Numerical evidence against a disproof). Exact integer/rational computation through $n = 6000$ gives

$$\max_{n \leq N} \log G_n \approx 0.75(\log N)^2;$$

over three decades the observed ratio to $(\log N)^2$ stays between about 0.6 and 0.9. The worst cases are diffuse rather than single-prime spikes. At the recorded worst indices the largest prime factor always occurs with $v_p = 1$, and no in-range full-multiplicity phantom occurs. This is the Poisson-independence scale predicted by the zero-count data and makes the available disproof mechanisms numerically nonviable. It is not a proof of the pointwise conjecture: the finite computation cannot exclude an extremely sparse later phase transition.

9. CONCLUSION

The density form of Problem 3.2 is unconditional. The reflection functional equation, the gap-eliminant theorem for U_g , the cross-arc lemma, and the structured-resonance theorem are unconditional as well. The only delicate dyadic blocks are in range, and there the reduction is

$$\begin{aligned} \text{terminal exact-collision conjecture} &\implies (P_2)^G \\ &\implies \sup K_X \text{ has a power saving} \\ &\implies \text{pointwise Problem 3.2.} \end{aligned}$$

Under the mass conjecture (C1), the terminal conjecture is equivalent, after renaming positive exponents, to the uniform single-arc cap (SA), the middle-prime cap (Ω), and the power-saving supremum target. The present argument does not claim that the qualitative conclusion $\log G_n = o(n)$ conversely forces a fixed power saving.

The terminal problem is a holonomic shifted-prime divisor problem in the circle of Hooley and Murty–Rosen–Silverman, sharpened to exact collisions. Lemma 14 shows how a compressive multiplicative invariant solves the Mersenne analogue; no counterpart is known for Apéry zeros. Phantom locality, the bandwidth model R9.1, and the Lang–Trotter implication R9.2 certify the limits of the currently banked divisibility and equidistribution routes. What remains is genuinely scale zero: either find a cross-prime compressive invariant, or prove an exact-collision theorem for holonomic values.

The six-fold map sharpens this status. The density theorem is fully unconditional. At the fixed-power endpoint, and under the mass conjecture (C1) where required, (SA), (Ω), and (NC) are equivalent after exponent renaming, and each implies pointwise Problem 3.2. This is the named Murty–Rosen–Silverman holonomic-sibling exact-collision problem. The computations in the appendix are compatible with the Poisson model and leave a large numerical margin. As stated above, no converse from the merely qualitative conclusion to a fixed power saving is claimed.

The reduction is optimal relative to the audited technology: five ensemble families—analytic, algebraic-geometric, Diophantine, dynamical, and additive combinatorics—share the phantom-transparency principle. The single-index adelic family instead stops at the critical ledger. The two certificates compose without circularity. Heights, separate per-place information, and ensemble statistics exhaust the input formats of the present multi-place tools, in the qualified sense of Remark 16.

A fully unconditional pointwise theorem therefore needs genuinely new place-mixing input of the type described there: a product formula for periods comparing the Betti/de Rham and crystalline realizations of the Apéry motive at one integer point. Standard GRH-type hypotheses do not formulate this comparison, and the valuation-one radical target is transparent to *abc*. The paper thus resolves the status of the pointwise problem while leaving its named scale-zero conjecture open.

The construction audit in Proposition 12 refines this last description. The elliptic-divisibility calibration shows that a product formula for periods would provide the right place-by-place *format*, but would not by itself locate the window-prime support. The exact fixed-power reduction is

$$\text{pointwise Problem 3.2} \iff (\text{SA}) \iff (\Omega) \iff (\text{NC}),$$

where the equivalences use (C1) when required and allow renaming the saving exponent. The reverse implication from the qualitative pointwise statement is not claimed.

The optimality certificate is therefore seven-fold: the five ensemble families, the single-index adelic family, and the motivic-comparison/construction gate R14. It is summarized by phantom-transparency, the critical adelic ledger, and quotient-free locating. Exact computation through $n = 6000$ supports, over the tested range, $\max_{n \leq N} \log G_n \approx 0.75(\log N)^2$ and exhibits no full-multiplicity phantom, but remains numerical evidence only. The missing mathematics is Conjecture 8: an effective locating theorem for entropy-positive self-characteristic local loci without a fixed quotient. None of the audited consequences of p -adic Hodge theory, GRH, *abc*, Vojta’s conjecture, or unlikely-intersection technology supplies it. In this precise sense the required theorem is new, rather than a new combination of the seven audited families.

APPENDIX A. VERIFICATION SUMMARY

- Lemmas 1–5, Lemma 8, and Theorem B were checked exactly on all 8434 triples (p, q, r) with $7 \leq p \leq 200$, including the support equivalence against exact G_n . The zero-pair certificate was checked on all 232 consecutive zero pairs with $p < 3000$, and the evaluation identities and reflection functional equation were checked exactly through $r = 9$.
- Lemma 6 was checked for all 31 primes $7 \leq p \leq 139$.

- Lemma 7 was checked for $q = 2, 3$ and all $p \leq 139$, in addition to its role in the 8434-triple support check.
- After noncentral roots are grouped into reflection pairs, the Z_b data are close to a Poisson law of mean $1/2$; the maximum $|Z_p|$ is 16 among the 2704 primes up to 80000. The empirical mean of $|Z_p|$ is about 1; the value $1/8$ belongs to a different collision statistic and is not a zero-count mean. Thus the $p^{2/3}$ bound is far from tight.
- In the pointwise probe,

$$B(n) = \#\{\text{bad primes } p > \sqrt{n}\} \leq 5$$

for every $n \leq 10^4$.

- At five dyadic scales, the all-range and in-range data were

X	400	800	1600	3200	6400
$\max_{m < X^2} K_X(m)$	3	4	4	4	5
$P_{2,X}$	6	28	49	109	276
$\max_{y \in [2X, X \log X]} K_X(y)$	2	2	3	3	3

These are numerical checks only.

- The reflection functional equation had no failure for $p \leq 12809$; its termwise binomial proof was checked for $p \leq 809$.
- For $n = 250, 500, 1000, 2000, 3000, 4000$, respectively, the numbers of distinct prime divisors of b_n in $(\sqrt{n}, n]$ were

$$2, 3, 0, 0, 0, 2,$$

and the corresponding full-multiplicity logarithmic ratios were

$$0.030, 0.029, 0, 0, 0, 0.003.$$

The largest observed middle-prime multiplicity was 1.

- The eliminant identity was checked in 116 cases over primes from 101 through 1601, with no failure. For $g = 2, 3, 4, 5$, symbolic calculation gave degrees 3, 6, 9, 12 and coefficient heights $231, 2.6 \cdot 10^5, 6.6 \cdot 10^8$, and $3.6 \cdot 10^{12}$.
- The $\zeta(2)$ central congruence had no mismatch for primes $5 \leq p \leq 1200$. At $X = 1600$, its out-of-range count was $C_{X,0} = 3934$, while six sampled in-range counts were 35, 26, 34, 38, 40, 32, on the random-model pair scale. These values support only the range correction, not an anti-collision theorem.
- Exact bignum computation of G_n through $n = 6000$ found, over three decades,

$$0.6 \lesssim \frac{\max_{n \leq N} \log G_n}{(\log N)^2} \lesssim 0.9.$$

At the recorded worst-ratio indices $n = 523, 1667, 2098, 3471, 4855$, the gcd mass was diffuse and the largest prime factor had exponent 1; no in-range full-multiplicity phantom was observed. These are numerical checks only.

APPENDIX B. v3 CHANGELOG

- Replaced the obsolete depth-3, BAC, singleton-row, and interval-overlap endpoints by the in-range depth-2 pair gate and the terminal single-arc exact-collision conjecture.
- Added exceptional-prime tolerance, the precise arc digit restriction, the (SA)–(Ω) comparison, and the exact identity (6.10). Power-saving equivalences now explicitly allow exponent renaming.
- Added the cross-arc lemma, the structured-resonance theorem, and the Apéry–Hasse and projective-Frobenius descriptions of Z_p .
- Added the Mersenne fibration calibration and barriers R9.1 and R9.2. R9.1 is labelled as a model because the source notes do not give an actual root-set realization.
- Corrected the zero-count mean to approximately 1, retained the Rhin–Viola constant $5.24673\dots$, restricted Lemma 14 to the admissible window, and removed the false fixed-exponent and exceptional- y equivalences.

APPENDIX C. v4 CHANGELOG

Added the sieve-complexity invariant, the complete six-fold barrier map and adelic ledger, the product-formula-for-periods target, and the corresponding pointwise-status conclusion.

APPENDIX D. v5 CHANGELOG

Added the R14 construction gate, EDS calibration, quotient-free-locating frontier, seven-fold barrier map, disproof probe, and refined pointwise-status conclusion.

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